DEPARTMENT OF
COMPUTER SCIENCE

FLOWCHARTS II

DR JEREMIAH O. BANDELE
PhD Electrical and Electronic Engineering
University of Nottingham

Learning objectives

At the end of this lesson, you should be able to:

- Explain some uses of flowcharts.
- Discuss various types of flowcharts.
- Discuss flowchart advantages, disadvantages and principles.

Flowchart uses

Some of the common uses of flowcharts include:

- To plan a new project.
- To document a project.
- To model a business process.
- To manage workflow.
- To manage data.
- To map computer algorithm.

Flowchart Types

There are various types of flowcharts but we will focus on the four main types:

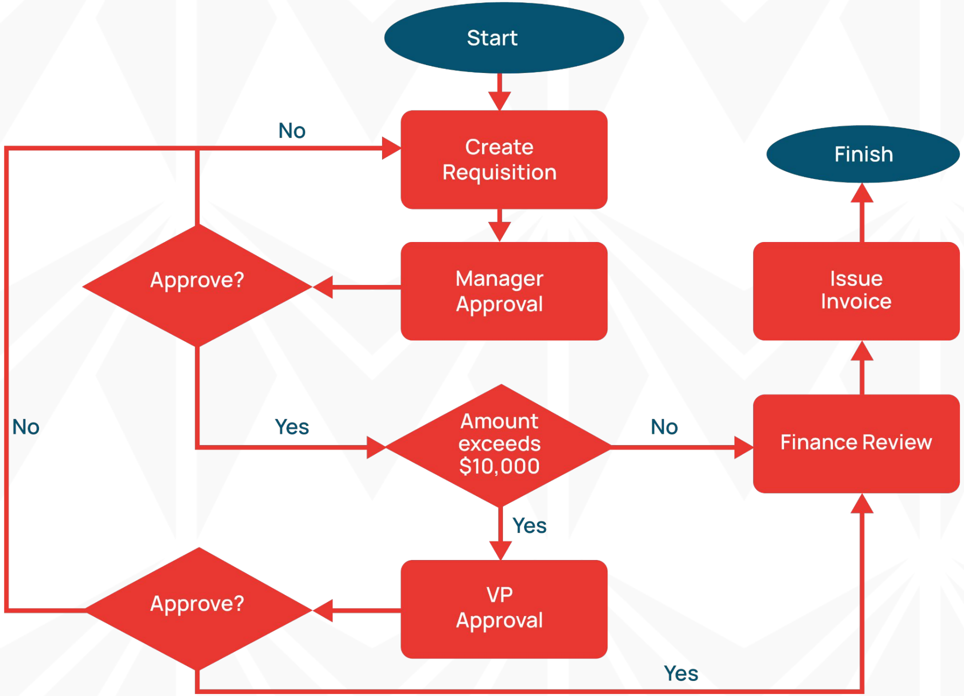
- Process flowchart
- Workflow chart
- Swimlane flowchart
- Data flowchart

Process flowchart

A process flowchart can be applied to almost anything and it can quickly describe how a process can be completed in an organisation. It can be used to:

- Clarify the responsibilities and roles within an organisation.
- Show the process a product goes through along the production chain.
- Describes information communication within an organisation.

Process flowchart: example



Workflow flowchart

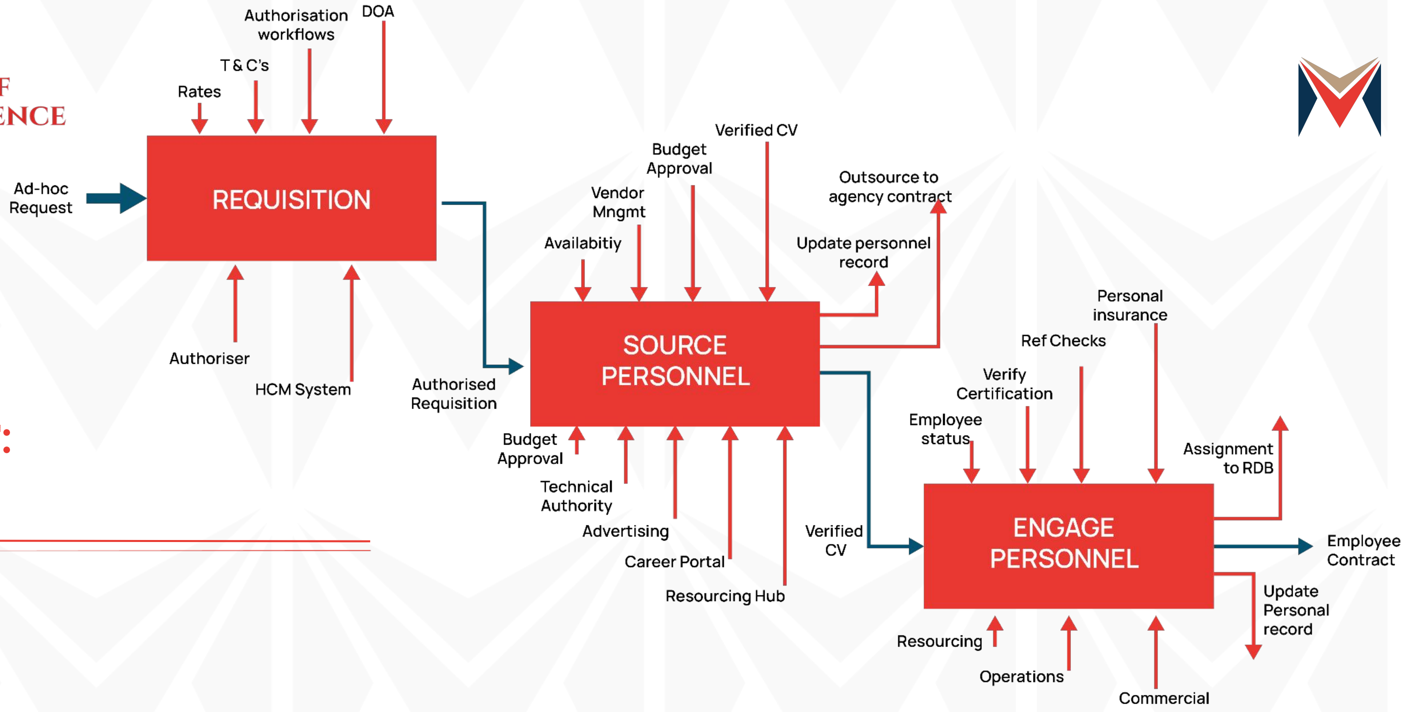
A workflow flowchart describes the manner in which businesses function.

It can be used:

- For employees training.
- To identify possible problematic areas.
- Simplify the operations of a business.
- To create an output that is stable and of a high quality based on standard procedures.

Department of Computer Science

Workflow flowchart: example

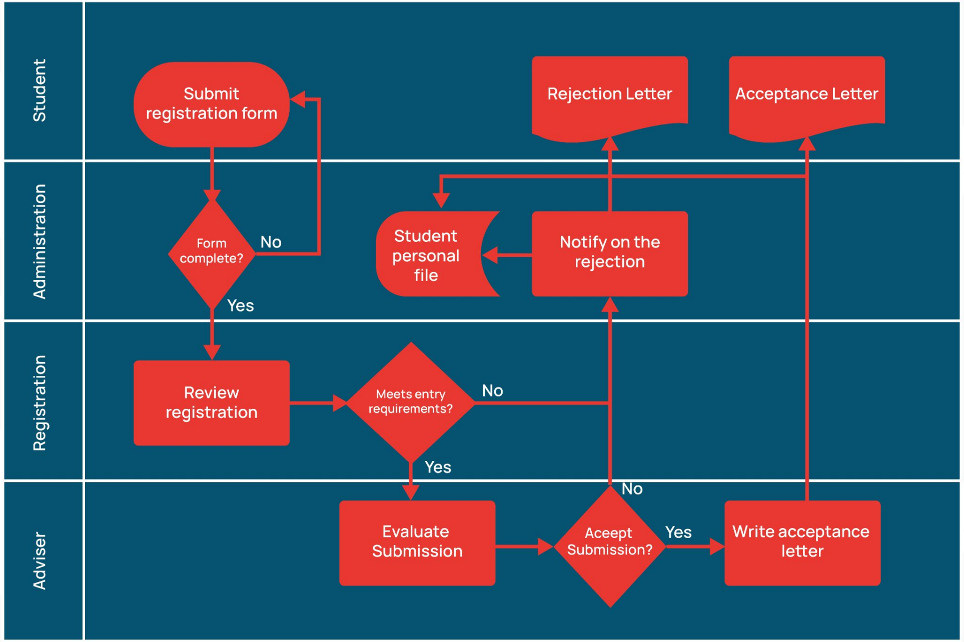


Swimlane flowchart

A swimlane flowchart is useful in applications where we have multiple information flowing side by side and we need to show it.

- It can be confused with a workflow flowchart but swimlane flowcharts allows for the creation of multiple categories of activities.
- It is ideal for representing the interaction of a process with different aspects of an organisation.

Swimlane flowchart: example



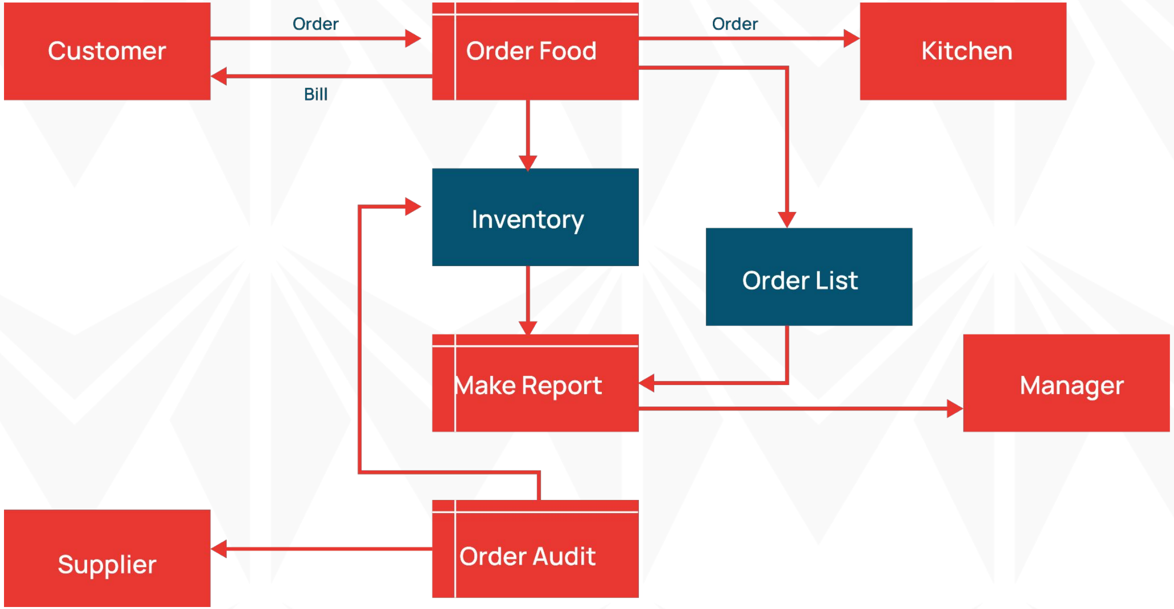
DATA FLOWCHART

A data flowchart describes the data processing aspect of the design and analysis of a system by showing the flow of information.

- Data flowcharts also ensure that the communication gap between the users and developers of the system is reduced.
- Data flowcharts can be used for the analysis of any form of information flow.

DEPARTMENT OF COMPUTER SCIENCE

DATA FLOWCHART: EXAMPLE



WHY WE NEED FLOWCHARTS

Flowcharts are needed for various reasons, let's see some of them:

- They are used to provide clarity to program logic.
- They are used to evaluate the actions resulting from setting some conditions.
- They are used to clarity a program's procedural steps.
- They are used to communicate the details of a program or process with other stakeholders.

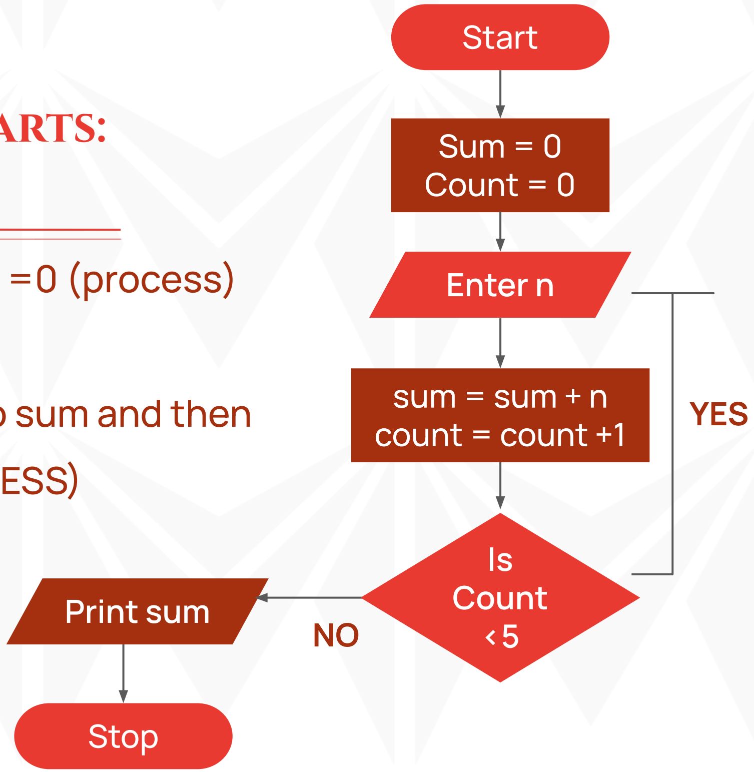
Why we need flowcharts: Example

- Initialize sum = 0 and count =0 (process)
- Enter n (I/O)
- Find sum + n and assign it to sum and then increment count by 1(PROCESS)
- Is count < 5 (DECISION)

  If YES go to step 2

  Else

  Print sum (I/O)



MID-LESSON QUESTIONS

1. A ____________ flowchart ensures that the communication gap between the users and developers of the system is reduced.
   a) Process flowchart
   b) Workflow chart
   c) Swimlane flowchart
   d) Data flowchart

Flowcharts advantages

There are some advantages to using flowcharts, let's see some of them:

- Flowcharts ensure that the logic of a process or program is simplified.
- Flowcharts are used to ascertain if the logic of the program works or not.
- Flowcharts are added to the program specification documents such that other programmers can easily understand the logic of the program.

Flowcharts disadvantages

There are some disadvantages to using flowcharts, let's see some of them:

- Some complex programs or processes might require that the flowchart spans multiple pages.
- Adjustments to the flowchart might be complicated and they might even require a new flowchart is drawn.
- The flowchart logic is hardly the most efficient when used to write a program because it can make the program unnecessarily longer than normal.

Flowcharts principles

When drawing a flowchart, it is necessary to be guided by some principles, let's see some of them:

- The flowchart should have a start point and a stop point.
- Flow lines should generally not cross, if possible.
- Instructions for comparisons should be made as simple as possible.
- The flowchart should be easy to follow, neat and clear such that the visual impact is very good.

FURTHER READING RESOURCES

- Chaudhuri, A. B. (2020). Flowchart and algorithm basics: The art of programming. Mercury Learning and Information.
- Chaudhuri, A. B. (2005). The Art of Programming Through Flowcharts & Algorithms. Firewall Media.
- Your Guide to the Most Common Types of Flowcharts. https://www.gliffy.com/blog/your-guide-to-the-most-common-types-of-flowcharts.
- Flowchart types. https://www.smartdraw.com/flowchart/flowchart-types.htm.
- Workflow Diagrams and Charts. https://www.integrify.com/workflow-diagram/.

REFERENCES

- Flowchart types. https://www.smartdraw.com/flowchart/flowchart-types.htm.
- Workflow Diagrams and Charts. https://www.integrify.com/workflow-diagram/.

Thank You